



microring modulators.



contents

Ring Resonator Simulation

- Lateral PN junction.
- Vertical PN junction.

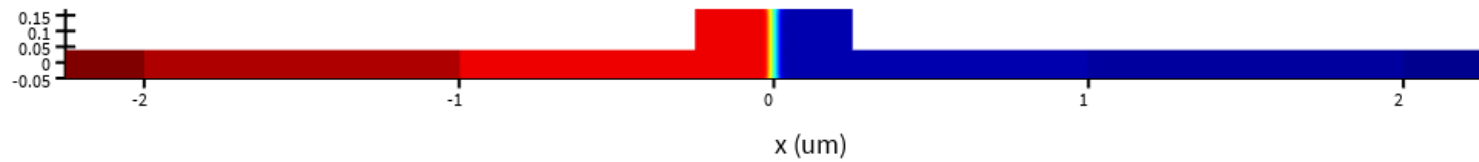
Documentation



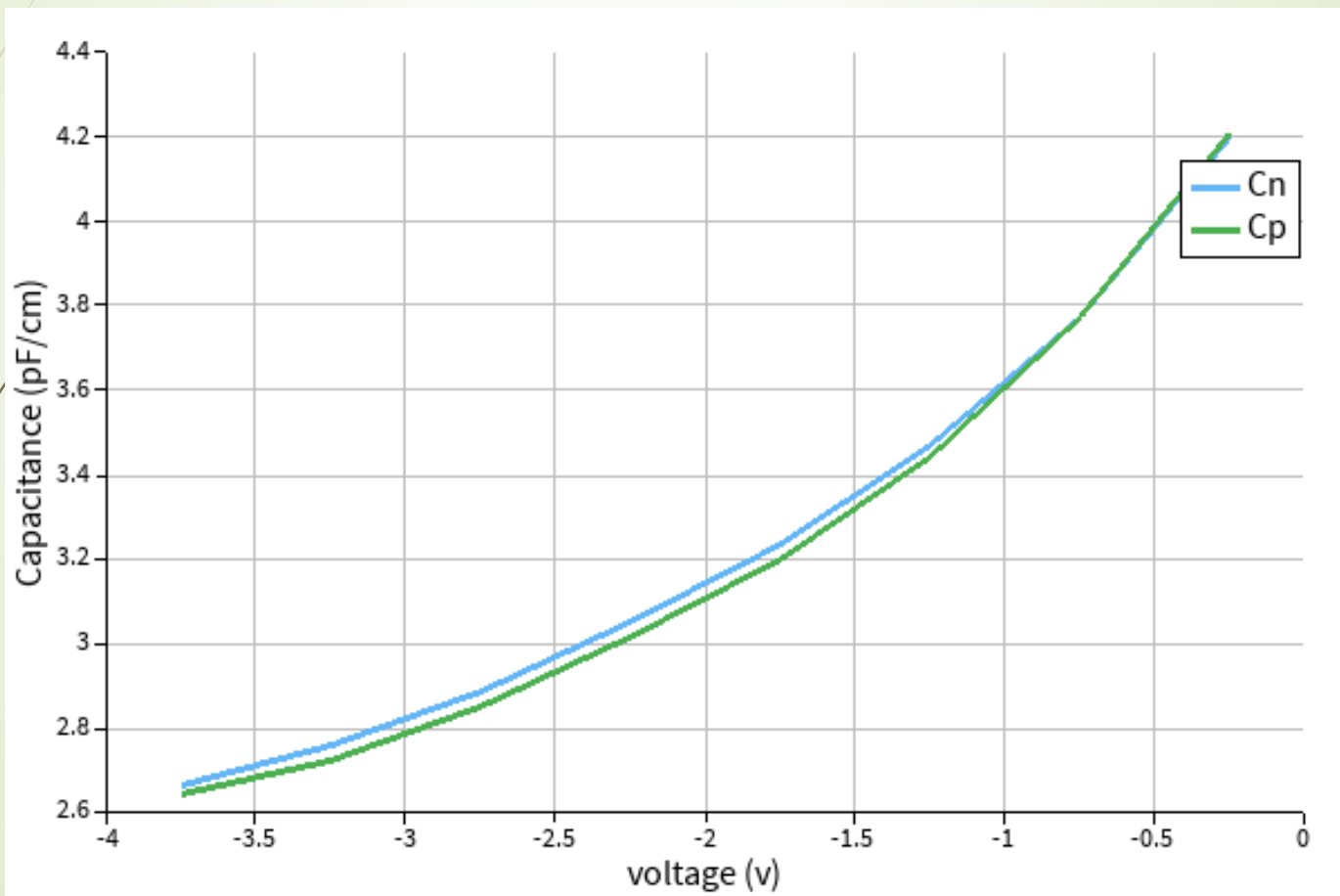
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- ➡ Lateral PN junction.

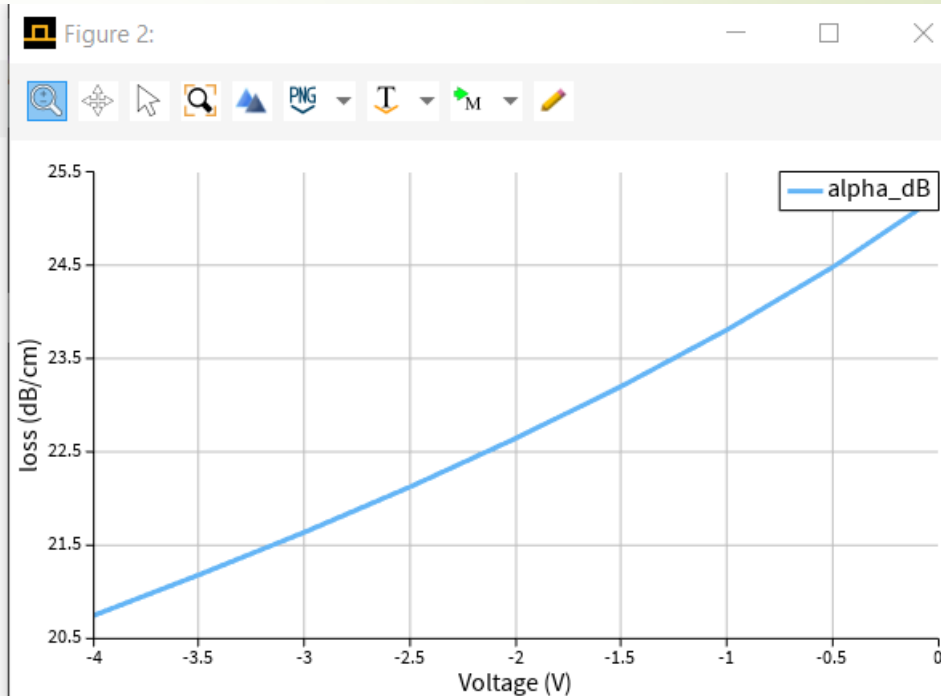
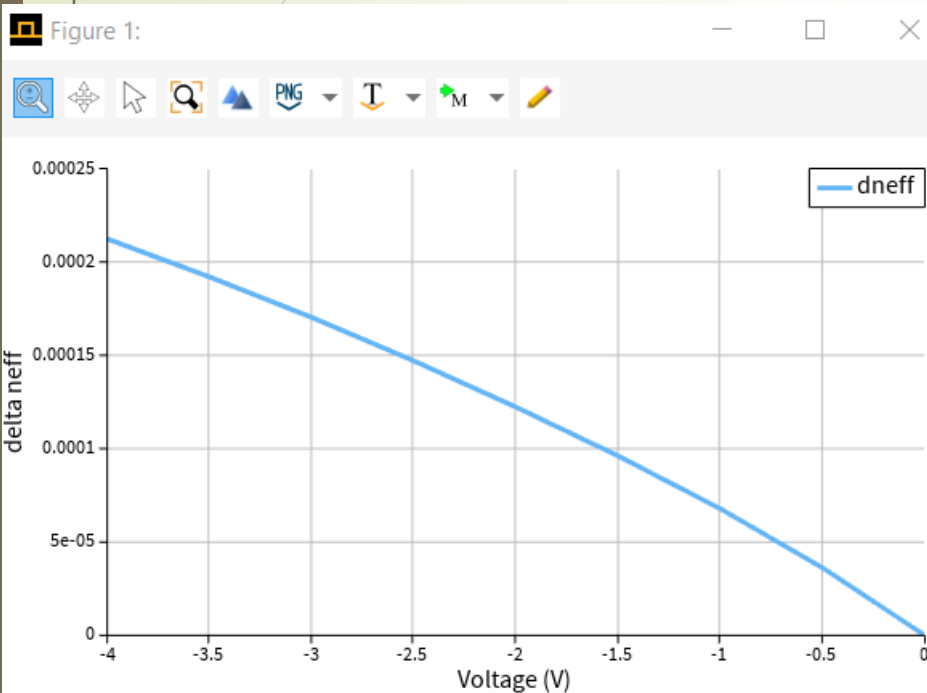
Lateral PN junction Simulation



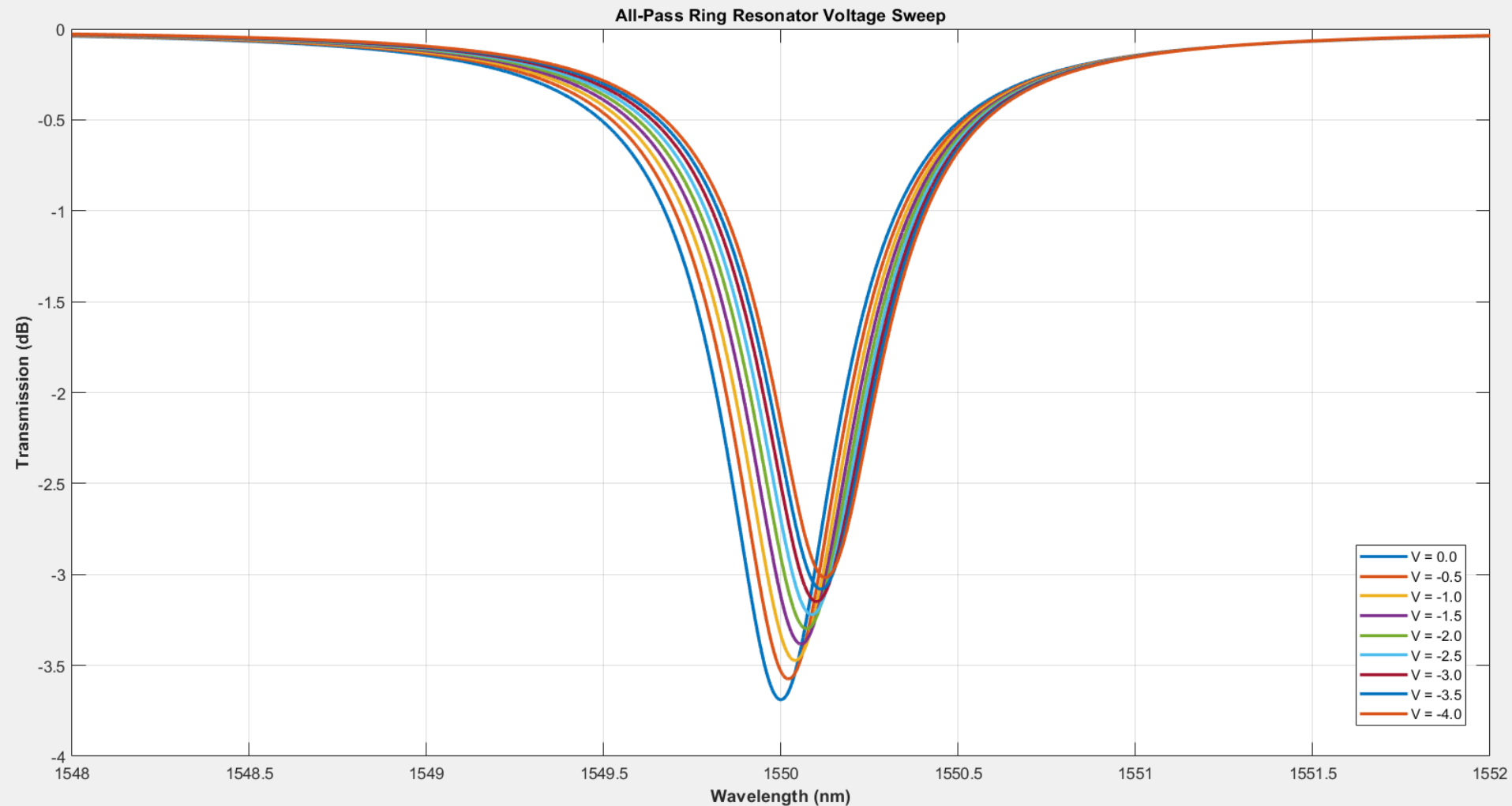
➤ Lateral PN junction Simulation



Lateral PN junction Simulation



➤ Lateral PN junction used for modulation

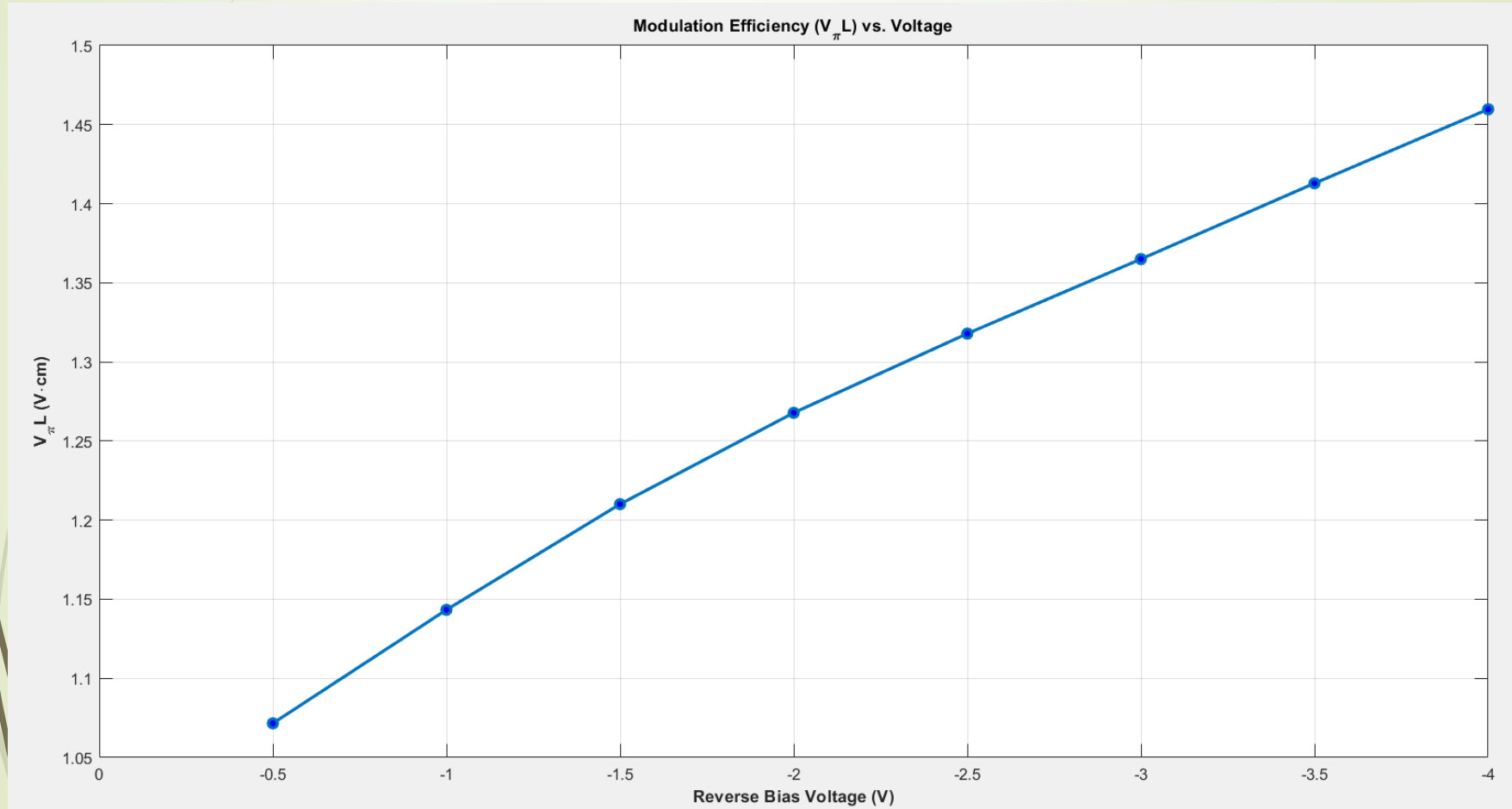




➤ Lateral PN junction used for modulation

- **FSR ≈ 9.86 nm**
- **FWHM ≈ 0.32 nm**
- **Q Factor ≈ 4837**
- **3dB BW ≈ 40 GHz**

➤ Lateral PN junction used for modulation





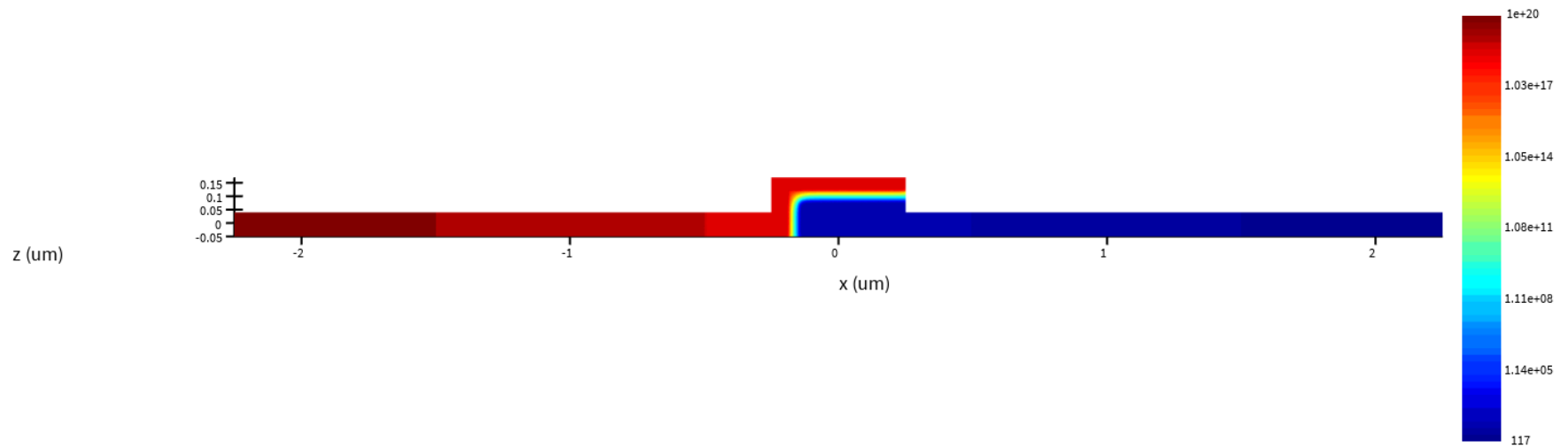
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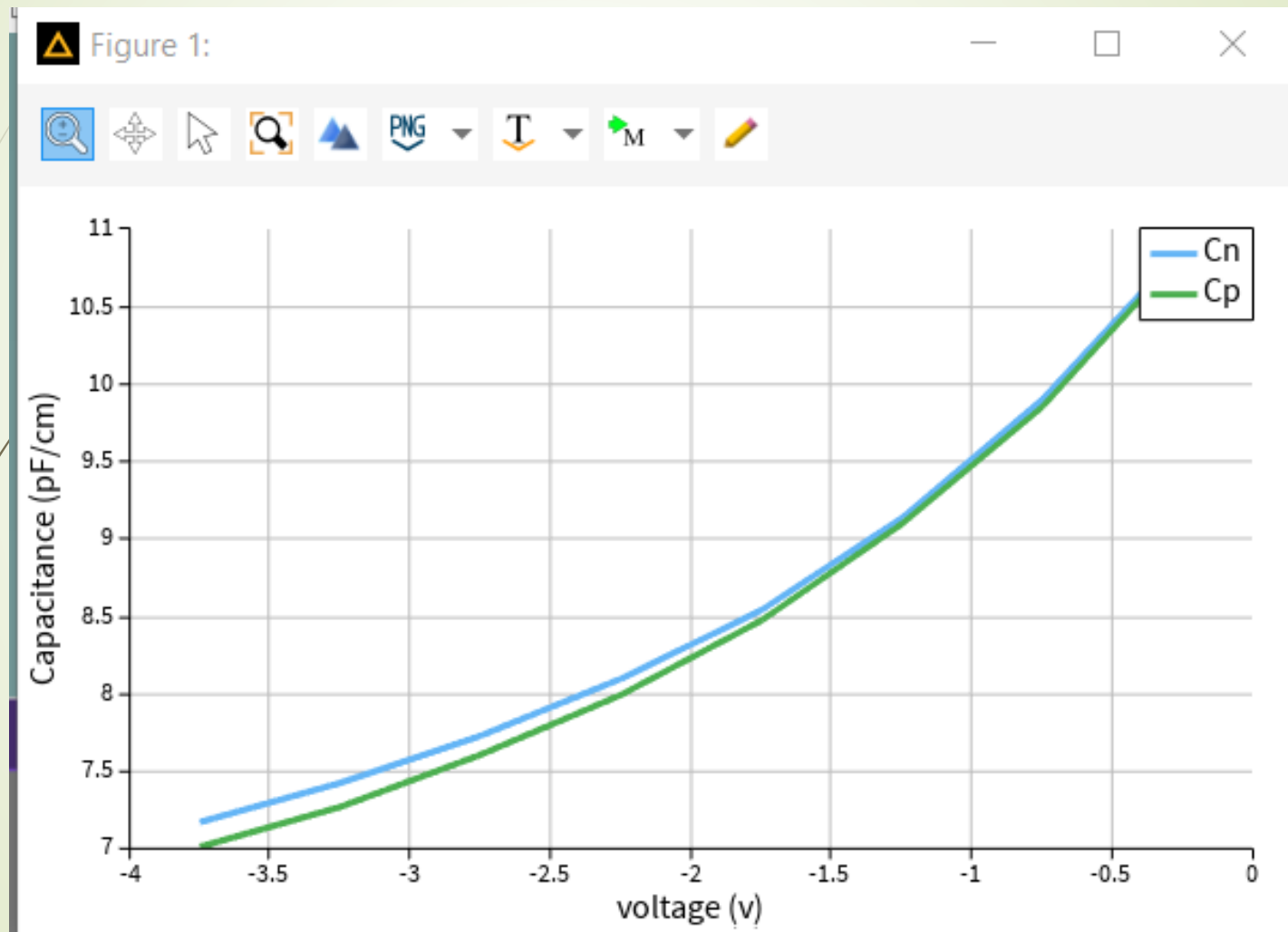
Vertical PN junction.



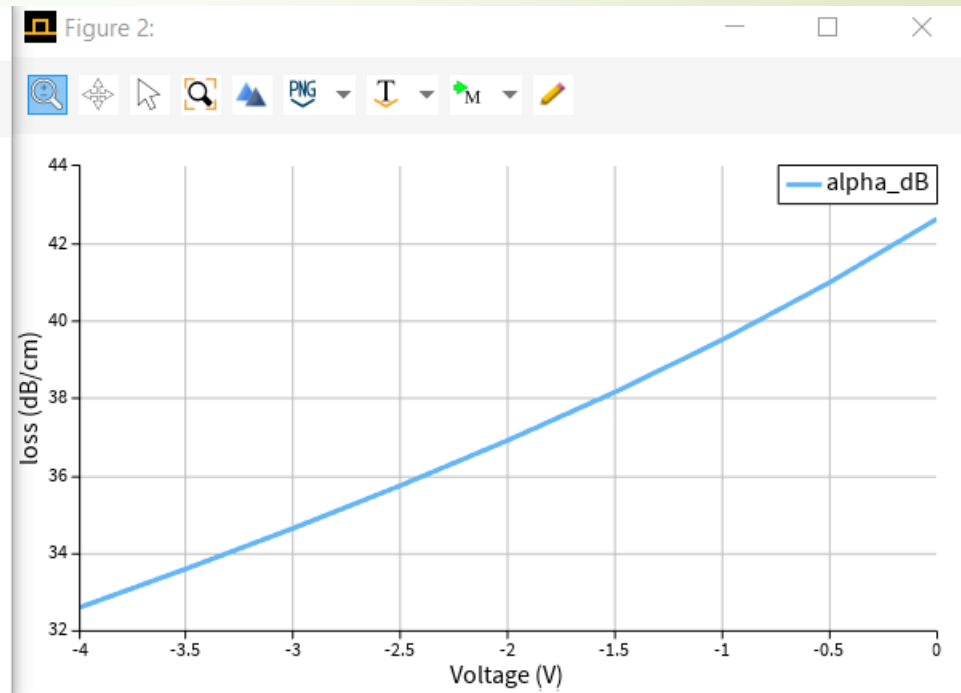
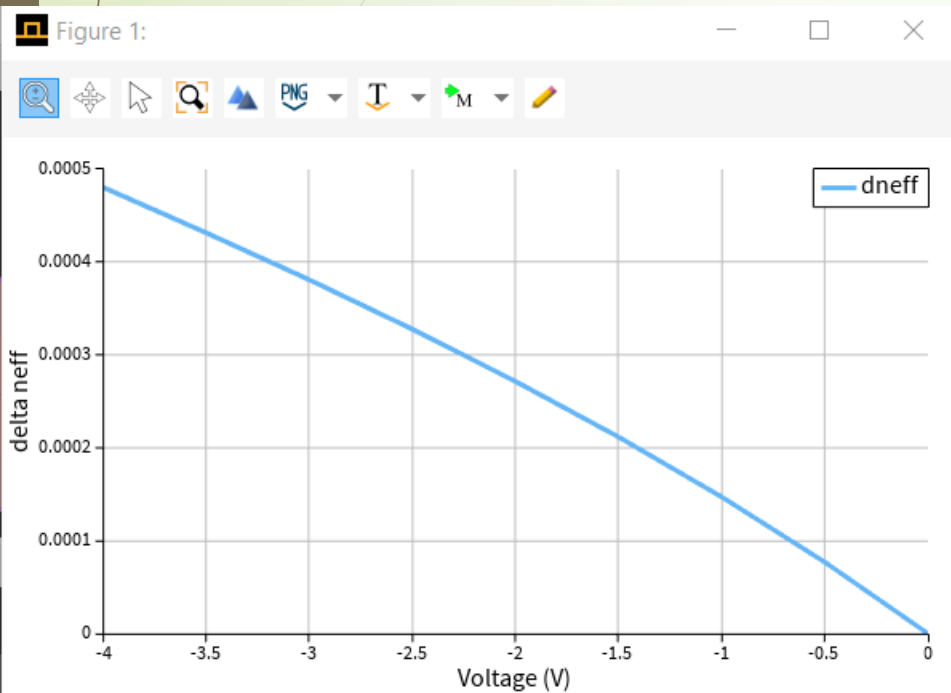
Vertical PN junction simulation



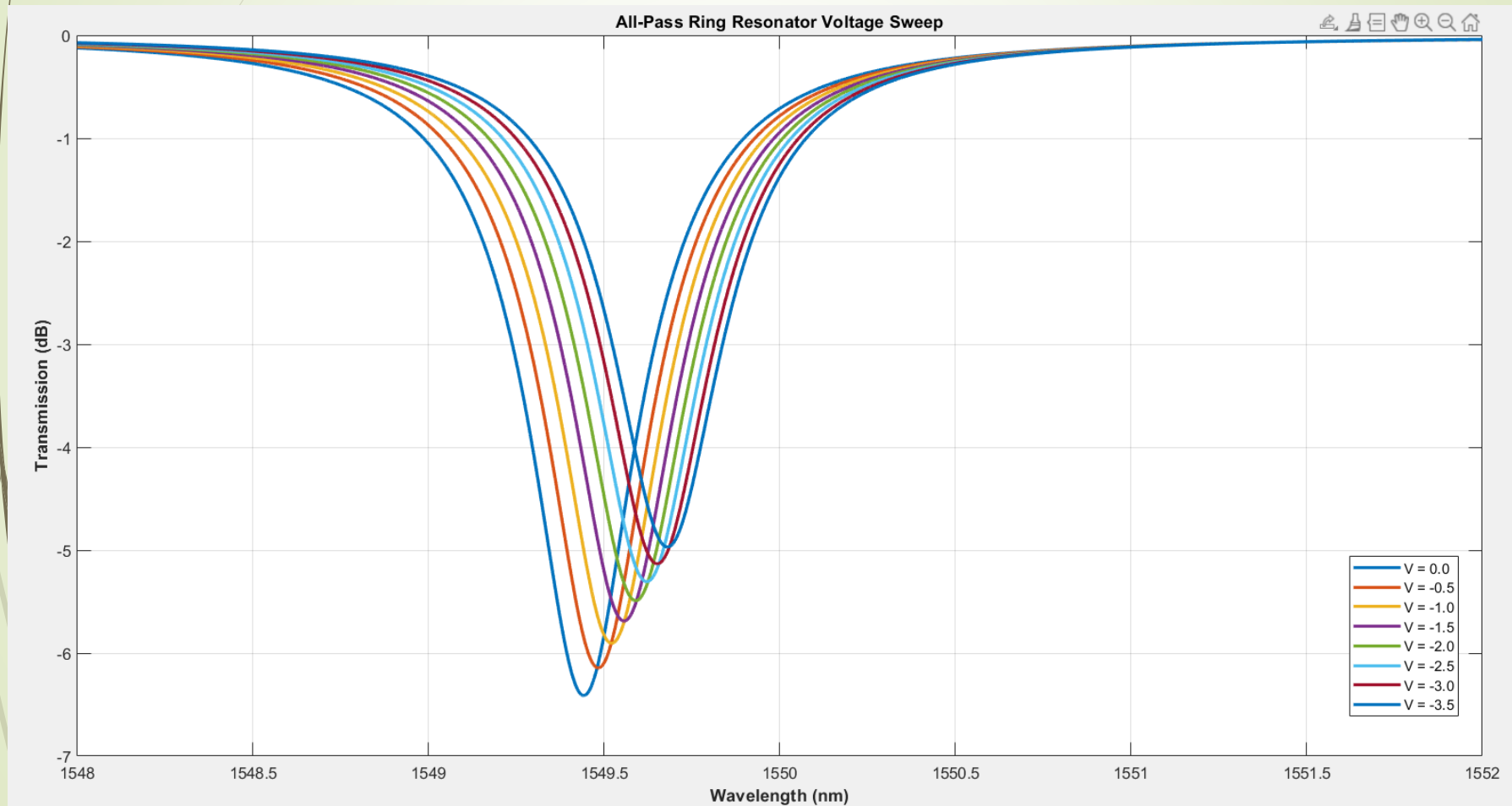
Vertical PN junction simulation



Vertical PN junction simulation



Vertical PN junction used for modulation

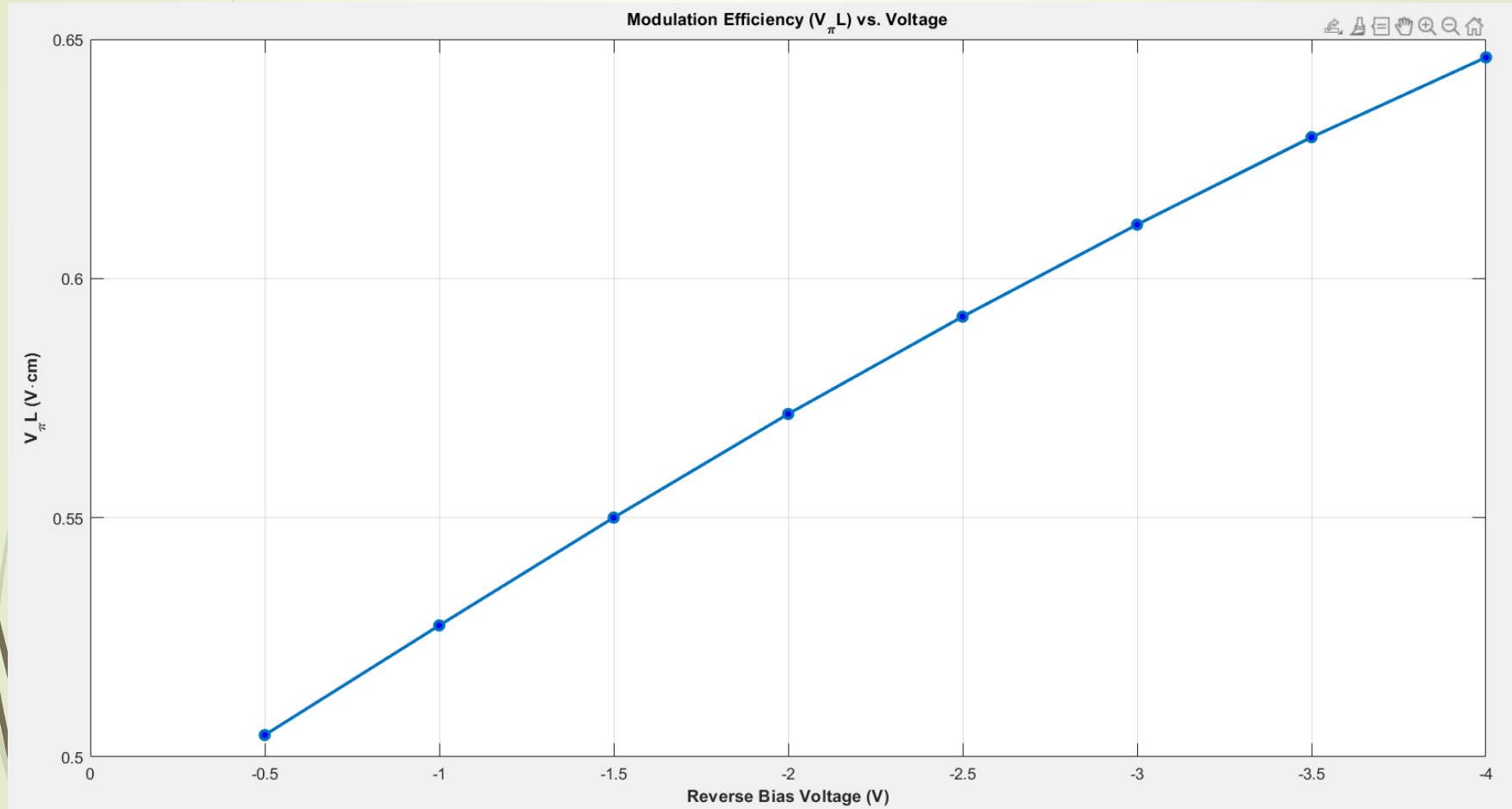




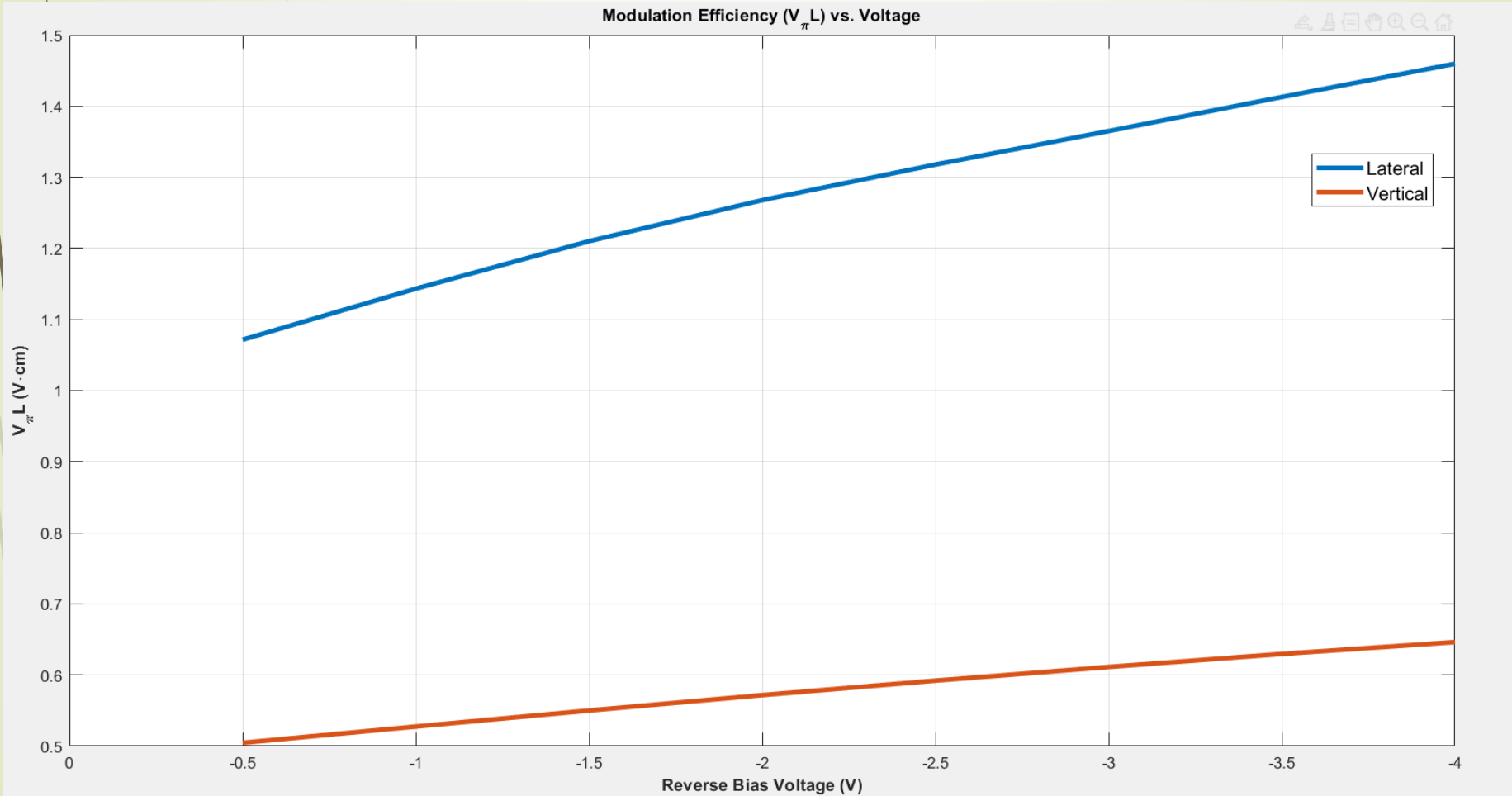
➤ Vertical PN junction used for modulation

- **FSR ≈ 9.85 nm**
- **FWHM ≈ 0.35 nm**
- **Q Factor ≈ 4465**
- **3dB BW ≈ 44 GHz**

Vertical PN junction used for modulation



Modulation efficiency Comparison

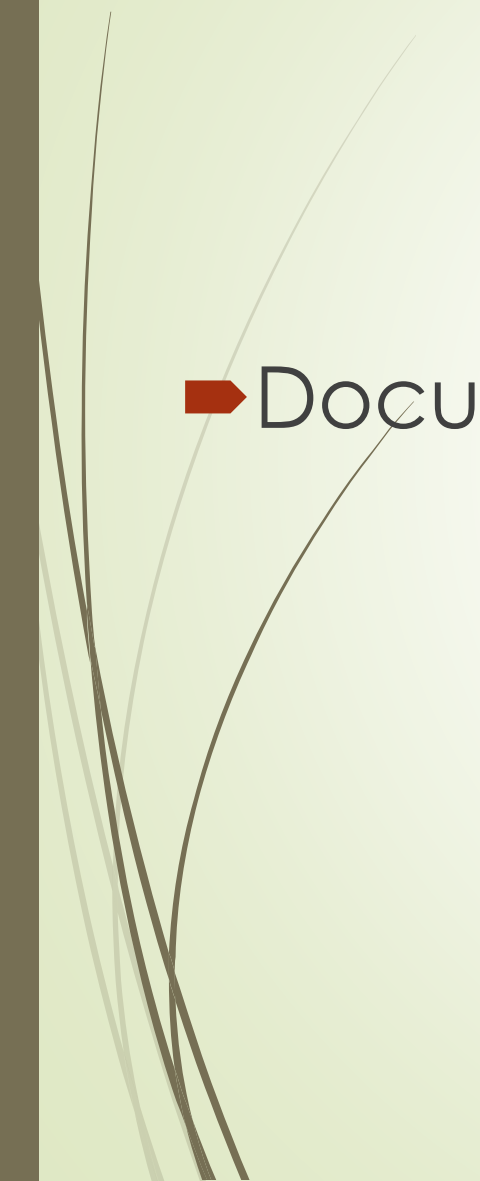


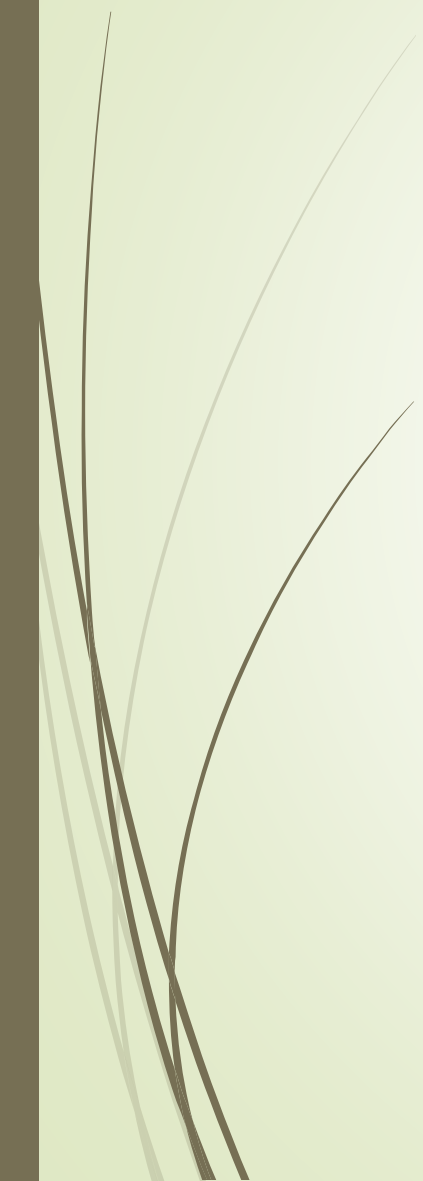


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Documentation.





Thanks 😊